

Part 2 report on the traffic analytics pipeline

Objective

Part 2 develops a reproducible Python workflow that transforms the raw Metro Interstate Traffic Volume CSV into cleaned and feature-engineered datasets for traffic analysis and subsequent machine-learning tasks. The workflow also generates visualisations of important traffic patterns and provides a command-line application for querying the processed data. Structured logging is incorporated throughout the workflow to record processing stages, data-quality interventions, generated outputs and invalid user inputs.

Methodology

Stage	Method	Result
Load & validate	Load CSV; validate schema and dates	48,204 rows, 9 columns
Standardise	Standardise categorical fields	48,143 non-holiday entries retained as missing
Deduplicate	Remove exact duplicates; retain repeated timestamps	17 removed; 48,187 rows remain
Validate numeric data	Flag temperature ≤ 0 K and rainfall > 500 mm/h	10 temperature and 1 rainfall value
Impute	Monthly median for temperature; overall median for rainfall	11 invalid values imputed
Save	Export cleaned dataset	48,187 rows, 9 columns

Exact duplicates were removed, while repeated timestamps were retained because they can represent multiple weather conditions within the same hour. Physically invalid temperatures (≤ 0 K) and implausible rainfall (> 500 mm/hour) were treated as missing and imputed using median values. Retaining repeated timestamps until feature engineering was important because removing them during cleaning would have discarded valid information about multiple weather conditions recorded within the same hour.

Feature engineering

`feature_engineering.py` consolidates the 48,187 cleaned records into 40,575 unique hourly observations and expands the dataset from 9 to 36 variables. Time features include hour, day of week, month, year, weekend/day-type indicators, traffic period, and sine and cosine cyclical

encodings of both hour and day of week. A whole-day holiday indicator is derived from the recorded holiday dates. Eleven binary weather indicators preserve the weather conditions reported at each timestamp, while temperature, cloud coverage and traffic volume are also standardised for subsequent analysis.

A four-level congestion target was derived from traffic-volume quartiles: Low ($\leq 1,248.5$ vehicles), Medium ($>1,248.5$ to $\leq 3,427$), High ($>3,427$ to $\leq 4,952$) and Severe ($>4,952$). This produced 10,144 Low, 10,144 Medium, 10,145 High and 10,142 Severe observations.

These engineered features were selected to represent different aspects of traffic behaviour. Hour, day-of-week and day-type variables capture recurring temporal and commuting patterns, while sine and cosine cyclical encodings of both hour and day of week represent their periodic nature without treating adjacent periods across cycle boundaries as unrelated. A whole-day holiday indicator identifies observations occurring on recorded holiday dates. Weather indicators preserve the potentially overlapping conditions recorded at each timestamp, and standardised numerical variables place continuous features on comparable scales for subsequent machine-learning algorithms. The four-level congestion category provides a categorical representation of traffic intensity that can be used in subsequent machine-learning analyses.

Findings

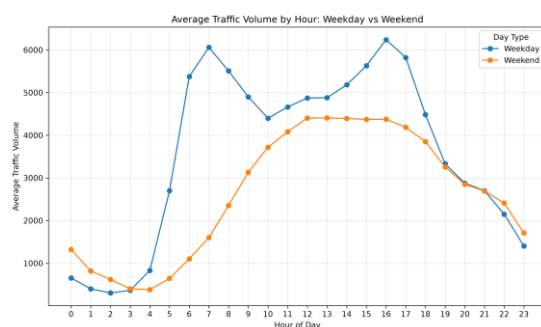


Figure 1. Average hourly traffic profile on weekdays and weekends

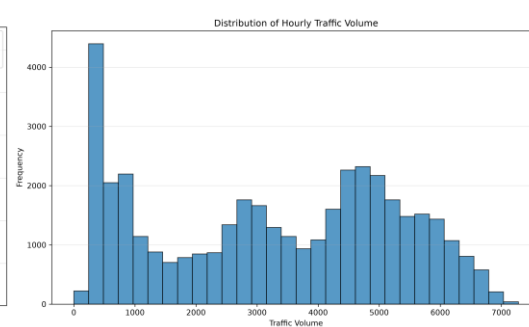


Figure 2. Distribution of hourly traffic volume

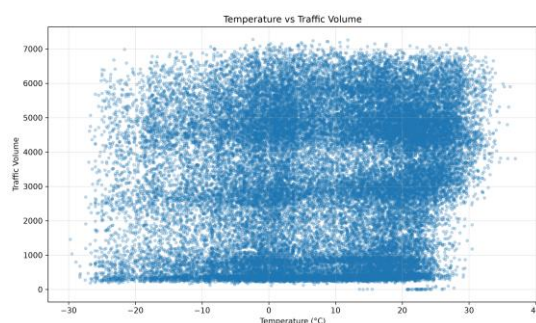


Figure 3. Temperature versus traffic volume

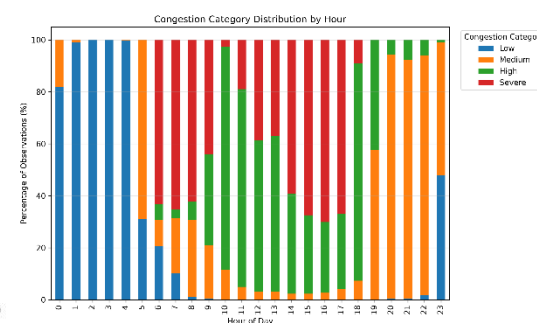


Figure 4. Congestion category by hour of day

Figure 1, hourly traffic profile. Weekday traffic shows pronounced commuting peaks, reaching approximately 6,241 vehicles/hour at 16:00, compared with a lower weekend maximum of approximately 4,408 at 13:00. This demonstrates the importance of hour and day type in predicting traffic demand.

Figure 2, traffic-volume distribution. Traffic volume ranges from 0 to 7,280 vehicles/hour, with a mean of 3,291 and median of 3,427. The multimodal distribution reflects distinct overnight, daytime and commuting traffic regimes.

Figure 3, temperature and traffic volume. Temperature has only a weak positive relationship with traffic volume ($r = 0.139$), with both high and low traffic occurring across a broad temperature range. Temperature alone therefore has limited explanatory value.

Figure 4, congestion by hour. Congestion varies strongly by time of day. Low congestion accounts for 100% of observations at 03:00, while Medium congestion is most prominent at 20:00 (approximately 93.9%), High congestion at 10:00 (approximately 85.8%), and Severe congestion at 16:00 (approximately 70.1%). This demonstrates a clear relationship between congestion level and hour of day and supports the importance of hour-based features for modelling traffic conditions.

Command-line application

`cli_app/traffic_cli.py` provides four commands: `date` summarises traffic for a selected date; `high-traffic` ranks hours by historical average traffic; `compares` weekday and weekend traffic; and `recommend` identifies lower-traffic hours within a specified time range. For example, weekday traffic averages 3,557 vehicles/hour, compared with 2,624 on weekends. Invalid inputs are handled with clear error messages and logged appropriately.

Logging and reproducibility

Each executable script uses `logging.getLogger(__name__)` with console and file handlers and records containing timestamp, level, module and message. INFO, WARNING, DEBUG and ERROR are used for normal milestones, recoverable issues, detailed troubleshooting and failures respectively; DEBUG feature-engineering output can be enabled with `--debug`. Internal status messages use logging rather than `print()`. Reproducibility is supported by `requirements.txt`, the documented execution order (`pipeline.py` → `feature_engineering.py` → `visualizations.py` → CLI), and the project README.

Limitations

- **Imputed sensor values:** Ten invalid temperature readings and one implausible rainfall reading were replaced with median values, so these represent estimates rather than observed measurements.
- **Dataset-relative congestion:** Low, Medium, High and Severe congestion are based on dataset quartiles rather than official traffic-engineering thresholds. The categories therefore describe relative congestion within this dataset and should not be interpreted as externally validated congestion classifications.
- **Repeated weather observations:** Multiple weather records at the same timestamp were consolidated into hourly binary indicators, reducing some detail from the original records.
- **Historical recommendations:** The CLI recommends lower-traffic periods using historical averages and does not account for real-time traffic conditions or incidents.